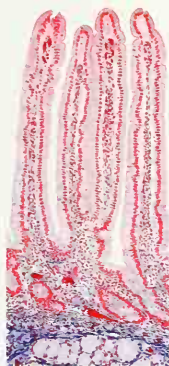
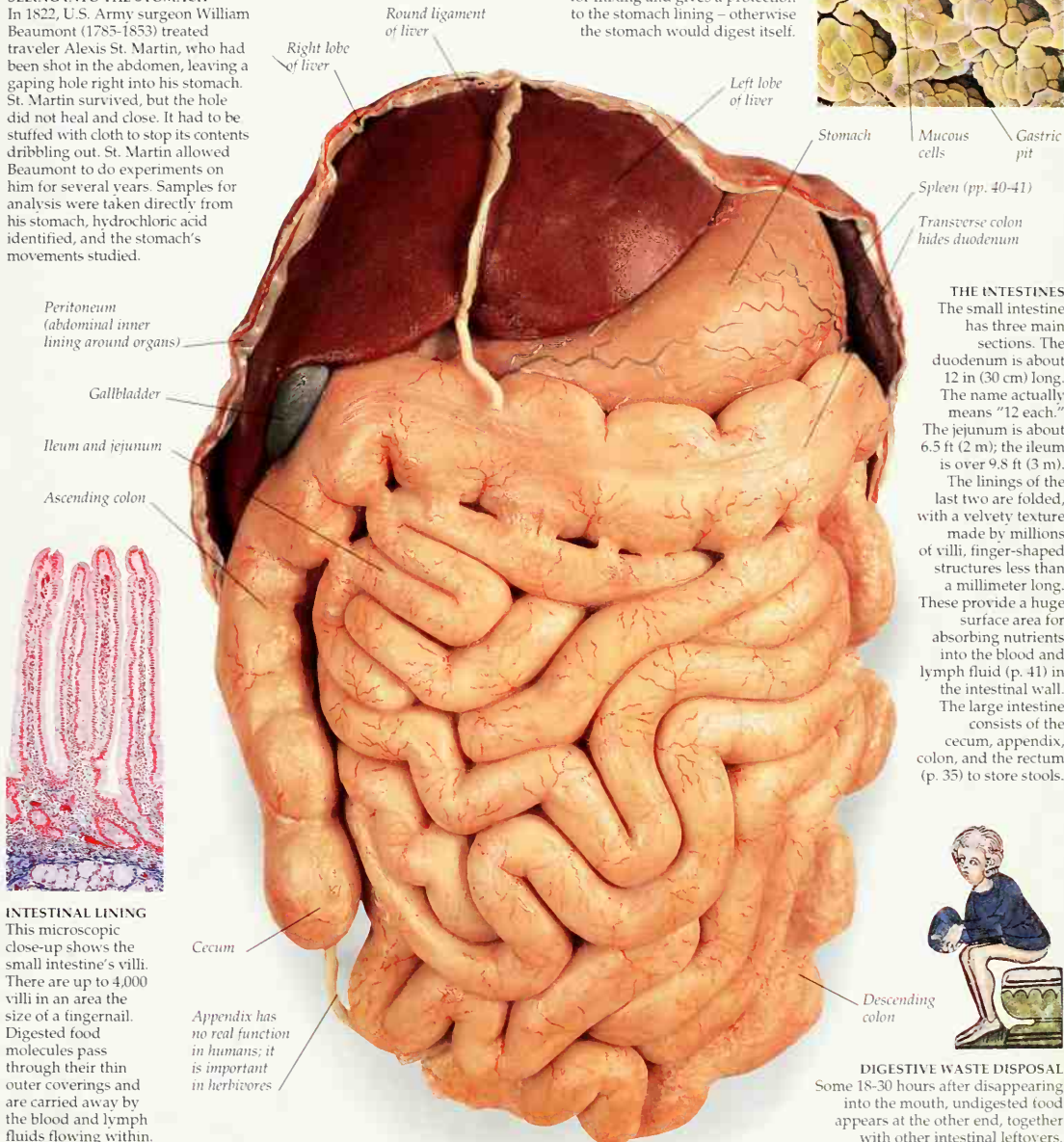


SEEING INTO THE STOMACH

In 1822, U.S. Army surgeon William Beaumont (1785-1853) treated traveler Alexis St. Martin, who had been shot in the abdomen, leaving a gaping hole right into his stomach. St. Martin survived, but the hole did not heal and close. It had to be stuffed with cloth to stop its contents dribbling out. St. Martin allowed Beaumont to do experiments on him for several years. Samples for analysis were taken directly from his stomach, hydrochloric acid identified, and the stomach's movements studied.

IN THE PIT OF THE STOMACH

The stomach's lining is dotted with thousands of tiny holes, each about 0.008 in (0.2 mm) across and 0.04 in (1 mm) deep, called gastric pits. The groups of cells lining the lower sides and bottoms of the pits are known as gastric glands. These cells make hydrochloric acid and digestive enzymes to attack food. Cells nearer the surface make thick, slimy mucus. This lubricates food for mixing and gives a protection to the stomach lining – otherwise the stomach would digest itself.



INTESTINAL LINING

This microscopic close-up shows the small intestine's villi. There are up to 4,000 villi in an area the size of a fingernail. Digested food molecules pass through their thin outer coverings and are carried away by the blood and lymph fluids flowing within.

THE INTESTINES

The small intestine has three main sections. The duodenum is about 12 in (30 cm) long. The name actually means "12 each." The jejunum is about 6.5 ft (2 m); the ileum is over 9.8 ft (3 m). The linings of the last two are folded, with a velvety texture made by millions of villi, finger-shaped structures less than a millimeter long. These provide a huge surface area for absorbing nutrients into the blood and lymph fluid (p. 41) in the intestinal wall. The large intestine consists of the cecum, appendix, colon, and the rectum (p. 35) to store stools.



DIGESTIVE WASTE DISPOSAL

Some 18-30 hours after disappearing into the mouth, undigested food appears at the other end, together with other intestinal leftovers.